



Aryan Baranwal

Roll No.:M25CSE035

Master of Technology (M.Tech)

Computer Science Engineering

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Github | LeetCode | GFG | Portfolio

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700+ DSA Problems Solved

EDUCATION

Degree/Certificate	Institute/Board	Year
M.Tech. (CSE)	Indian Institute of Technology, Jodhpur (IIT Jodhpur)	2025–2027
B.Tech. (ISE)	Jain University (Deemed-to-be University)	2020–2024

EXPERIENCE

- Image Analysis and Biometrics Lab (IAB), IIT Jodhpur** May 2025 – Present
Researcher – under the guidance of Dr. Richa Singh and Dr. Mayank Vatsa
Jodhpur, Rajasthan
– Developing and evaluating **5+ audio deepfake detection models** using **Python, PyTorch, and Transformers**, benchmarking **EER, ROC-AUC, and min-tDCF** across **3+ datasets**.
– Designing a **5-agent DARWIN-Voice framework** comprising **Generator, Detector, Attack, RAG, and Evolution agents**, integrating **LLMs, LangChain, and LangGraph** for adaptive audio deepfake detection.
- Nexus Institute of Information Technology Pvt. Ltd.** Jun. 2023 – Sep. 2023
Full Stack Developer Intern On-Site
– Developed and maintained **full-stack web applications** across production client projects using **AngularJS, React.js, Node.js, Express.js, and MongoDB**, integrating **RESTful APIs** for reliable frontend-backend communication.

PROJECTS

- WaveChat – Random Video Chat Platform** Jan. 2025 – May 2025
React, TypeScript, Python, FastAPI, WebRTC, WebSockets, Redis, Docker Github | Live URL | Video Demo
– Developed a **distributed real-time video chat platform** with decoupled Matching and Signalling microservices, backed by a **Redis queue**, with **STUN/TURN** integration and support for reliable peer-to-peer communication.
- EventFlow – Real-Time Event Processing System** Jan. 2026 – May 2026
Java, Spring Boot, Apache Kafka, PostgreSQL, Docker, Kubernetes, GCP Github | Live URL | Video Demo
– Architected a **real-time scalable event-driven microservices platform** using **Java, Spring Boot, Apache Kafka**, and **PostgreSQL**, containerized with **Docker** and orchestrated using **Kubernetes** and **HPA** with **GCP/GKE**.
- Multi-Agent Research Paper Reviewer** May. 2026 – Jun. 2026
LangGraph, MCP, LLMs, Ollama, Hugging Face, ArXiv API, PDF Processing, Streamlit Github | Live URL | Video Demo
– Engineered a **multi-agent research paper review system** using **LangGraph, MCP, and LLMs**, orchestrating specialized Reader, Critic, and Meta-Reviewer agents for automated **paper analysis, summarization, and validation**.
- Audio Deepfake Detection (XLS-R + SLS) – Paper Reproduction** Jan. 2026 – May 2026
Python, PyTorch, Hugging Face Transformers, XLS-R 300M, RawBoost Github | Live URL | Video Demo
– Reproduced the **XLS-R 300M + SLS** architecture for **audio deepfake detection**, evaluating ASVspoof 2021 DF/LA and In-The-Wild with **EER and min-tDCF**; ablating **fine-tuning, RawBoost augmentation, and layer selection**.

KEY COURSES TAKEN

- Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Advanced DSA, Software and Data Engineering, Virtualization and Cloud Computing, Operating Systems, Computer Architecture

TECHNICAL SKILLS

- Programming:** Java, Python, JavaScript, SQL, Object-Oriented Programming, Data Structures, Algorithms
- Backend & Systems:** Spring Boot, Node.js, Express.js, FastAPI, REST APIs, JWT, Socket.IO, WebSockets, WebRTC, Microservices, Distributed Systems, Scalable System Design, Real-Time Communication Systems
- Frontend:** React.js, AngularJS, HTML, CSS, Bootstrap, Material UI, Tailwind CSS, Responsive Design
- Databases & Messaging:** MongoDB, PostgreSQL, Redis, Apache Kafka, Elasticsearch, Firebase, Relational DB
- AI & Machine Learning:** PyTorch, Hugging Face Transformers, Scikit-learn, Deep Learning, Natural Language Processing, Computer Vision, LLMs, Generative AI, Agentic AI, Multi-Agent Systems
- Research & Applied AI:** RAG, LangGraph, LangChain, Model Context Protocol (MCP), Vector Search (FAISS), Model Evaluation & Benchmarking, Experiment Design, Statistical Analysis, Ollama, OpenAI API
- Data Science:** SQL, Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA), Data Visualization
- DevOps, Cloud & MLOps:** Docker, Kubernetes, MLflow, GitHub Actions, Prefect, Model Serving, CI/CD, GCP, Cloud Run, Artifact Registry, Minikube, Jenkins, Prometheus, Linux/Unix environments
- Tools:** Git, GitHub, Postman, IntelliJ IDEA, VS Code, Eclipse

ACHIEVEMENTS

- Solved 700+ Data Structures & Algorithms Problems** Across Leetcode and GeeksForGeeks 2026
- IndiaAI Fellowship** Selected for the national-level fellowship supporting advanced research and innovation in Artificial Intelligence. 2026
- 3rd Place Syntax Showdown** (Competitive Programming Contest), Sandstone Summit 5.0, IIT Jodhpur, in collaboration with GeeksforGeeks. 2025
- Smart India Hackathon (SIH)** Selected through institute-level screening. 2025
- Hacktoberfest Contributor** Contributed to open-source projects through accepted pull requests. 2022, 2023